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Is lower hip range of motion a risk factor for groin pain in athletes? A systematic review with clinical applications

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► Additional material is published online only. To view please visit the journal online (http://dx.doi.org/10.1136/bjsports-2016-096619).

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Accepted 27 February 2017
Published Online First 21 April 2017

ABSTRACT

Background Whether hip range of motion (ROM) is a risk factor for groin pain in athletes is not known.
Objectives To systematically review the relationship between hip ROM and groin pain in athletes in cross-sectional/case–control and prospective studies.
Study design Systematic review, prospectively registered (PROSPERO) according to PRISMA guidelines.
Methods Pubmed, Embase, CINAHL and SPORTDiscus were systematically searched up to December 2015. Two authors performed study selection, data extraction/analysis, quality assessment (Critical Appraisal Skills Programme) and strength of evidence synthesis.
Results We identified seven prospective and four case–control studies. The total quality score ranged from 29% to 92%. Heterogeneity in groin pain classification, injury definitions and physical assessment precluded data pooling. There was strong evidence that total rotation of both hips below 85° measured at the pre-season screening was a risk factor for groin pain development. Strong evidence suggested that internal rotation, abduction and extension were not associated with the risk or presence of groin pain.
Conclusion Total hip ROM is the factor most consistently related to groin pain in athletes. Screening for hip ROM is unlikely to correctly identify an athlete at risk of developing groin pain because of the small ROM differences found and poor ROM measurement properties.

INTRODUCTION

Groin pain is common in sports involving explosive movements, directional changes, repeated kicking and body contact.[1] The aetiology of groin pain is unclear and probably multifactorial.[2] Seven systematic reviews[3–9] have focused on factors associated with groin pain in athletes. Lower hip range of motion (ROM) was a risk factor in four systematic reviews,[4 7–9] but not in two others.[3 6] Mosler *et al*[5] studied factors that differentiated athletes with and without hip and groin pain and found lower hip ROM to be associated with its presence.

A recent international agreement reported that the hip can be an important cause of groin pain in athletes.[1] The interference of hip-related pathology, hip ROM and groin pain was not examined in these reviews. Additionally, there is no clear understanding of the paradigm of decreased hip ROM in relation to groin pain.[10 11]

No published prevention or treatment programmes for groin pain in athletes focus specifically on hip ROM.[1 3] A recent meta-analysis of seven randomised trials on preventing groin injuries in athletes showed a lack of efficacy.[12] These prevention programmes mainly consisted of active exercise and did not examine hip ROM at baseline or follow-up. A review appraising existing literature on hip ROM measures and their relation to groin pain would assist in the planning of new preventative strategies and making adequate choices in study designs.

The primary aim of this review was assess whether there was a relationship between hip ROM and groin pain in athletes. The secondary aim was to guide clinicians and researchers in interpreting the findings on the relationship between hip ROM and groin pain. The key research question was, is there a relationship between hip ROM and the presence of groin pain in athletes?

METHODS

This systematic review was conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) guidelines. The protocol for this review was registered at the PROSPERO register (website; http://www.crd.york.ac.uk/PROSPERO) for systematic reviews under registration number CRD42015017666.

Search strategy

Before this systematic review the Cochrane Database of Systematic Reviews, MEDLINE and PEDro were searched for systematic reviews on groin pain in athletes involving hip ROM to ensure that a similar study had not already been conducted. No date restrictions were applied and all databases were searched in full up to 1 August 2015 when this review process started.

To identify studies that met the inclusion criteria, a search was conducted on 1 December 2015 in the electronic databases Pubmed, Embase, CINAHL and SPORTDiscus without any date restriction. A combination of the following keywords and their synonyms was used: *groin pain*, *hip ROM*, *risk factors* (see online supplementary appendix 1 for the full search strategy). Cross-references from previous reviews and all retained articles that fulfilled the inclusion criteria were screened for possible relevance. Additionally, three experts in the field were asked whether they had any relevant references available from books or other sources. All

To cite: Tak I, Engelaar L, Gouttebarge V, *et al*. *Br J Sports Med* 2017;**51**:1611–1621.

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results were entered in Reference Manager (Thomson Reuters, Philadelphia, USA). Duplicates were removed.

Selection of studies

Two reviewers (LE and IT) independently reviewed all obtained titles and abstracts to determine possible eligibility. They checked full-text versions of all the retained articles and screened reference lists of these studies and the systematic reviews already identified, for other relevant citations. Full-text screening was then performed according to the eligibility criteria. The selection status of each retained article was discussed and disagreements were resolved by consensus. A third reviewer (AW) was available to discuss ongoing disagreement. A final list of articles for further detailed analysis was created.

Eligibility criteria

Studies were considered eligible for this review if data were presented on hip ROM in athletes in a study evaluating risk of developing groin pain (prospective longitudinal study), OR when comparing athletes with and without groin pain (cross-sectional/case series study). Groin pain could be described according to the numerous past different definitions in the literature.[1]

A study was only included if it fulfilled the following criteria:

a. an original study, published in English, German, French or Dutch;
b. an observational (prospective) cohort, cross-sectional or case–control study;
c. studying athletes as the population of interest;
d. studying groin pain as a variable;
e. studying hip ROM as a variable;
f. presenting statistical data analyses of the relationship between dependent and independent variables (groin pain and hip ROM).

This review aimed to assist clinicians working with athletes who might be at risk for, or have already developed, groin pain. Studies on hip and/or groin pain caused by known hip conditions were excluded as being outside the scope of this review. Intervention studies were excluded, as were studies on cadavers or animals.

Data extraction, synthesis and analysis

Two reviewers (LE and IT) extracted the following data, using a standardised form:

- definitions of groin pain and the accompanying injury criteria (used terms, specifications of groin pain and (physical) examination criteria);
- onset of pain;
- time of existing complaints;
- time loss;
- study setting (sports type and level, gender, country);
- study setting and design;
- details of ROM assessment (movement directions, body position, measurement device, reporting of data in appropriate Système International units,[13] definition of end ROM and the number and qualification of assessors).

The study results and outcomes are presented as the number of participants, injuries and (injured) hips, including percentage. It is reported whether injured players or injured hips were used for analyses and which data were provided (per side, for all hips or bilateral hips (BH) as sum scores). As a previous injury increases the risk of a re-injury, any reporting of re-injuries and methodological/statistical considerations (whether correction was needed or not) are presented. Data are analysed descriptively. For the studies that supplied adequate continuous data, means and SD or SE of means are presented. For dichotomous data, ORs/RR with matching 95% CIs are presented. Mean differences (MD) for each ROM movement are calculated and presented, rounded off to whole numbers as most measurement tools have 1° increments.

Risk of bias assessment

We used the Critical Appraisal Skills Programme (CASP, 2014, www.casp-uk.net) check list to assess the risk of bias in included studies, and specifically adapted[14 15] (LE, IT, MB) it for our systematic review. The criteria list contains items on information, validity and/or precision in the following categories: study objective, study population, exposure measurement, assessment and analysis of the outcome and data presentation (see online supplementary appendix 2 for the full criteria list).

Two independent reviewers (LE and IT) rated each study as 'positive' (+) if it included an item, 'negative' (−) if it did not include an item, 'non-applicable' (NA) if that item was not applicable to that particular study. When no clear information was present on this item a 'negative' (−) was scored. The reviewers compared their results and in cases of disagreement, consensus on each item was reached in a meeting. Absolute values and level of agreement (Cohen's κ) are presented. The number of items that were rated positively was divided by the total achievable (applicable items only) score to provide a 'quality score'. The studies were ranked as a percentage of the maximum achievable score, and, relative to this total score, categorised as scoring high or low according to the CASP criteria.[16] A high score, thus a low risk of bias study, was a study that scored positively for >50% of the validity/precision items on the methodological quality list. A low score, referring to a high risk of bias study, was one that scored positively for ≤50% of the validity/precision items.[14 17] Quality scores were obtained for each study type and overall. All studies were checked for reporting on conflicts of interest.[18]

Item assessment

Item assessment was performed by calculating the ratio of positively scored items/negatively scored items for each CASP item over all the studies. When no negative score was obtained a 'maximum' score was assigned and when no positive score was obtained a 'minimum' score was assigned. This was done in order to provide an insight into which items, in general, had low or high scores among the studies included.

Strength-of-evidence assessment

The strength of evidence for hip ROM as a potential risk factor for, or in association with, groin pain was assessed by defining four levels of evidence (see table 1).[17] A relationship was rated as positive when the risk estimate was increased. No relationship was identified when the effect estimate indicated no increased risk or a decreased risk of hip ROM or when it was reported to be statistically non-significant ($p>0.05$) without reporting the risk estimate. Significant differences were based on means with SD and/or matching 95% CI. If a study provided neither of these two, any information on significant differences ($p<0.05$) between the groups was used.

RESULTS

Search results of this study

The database search resulted in 784 studies of which 208 duplicates were removed. Screening of titles and abstracts was performed on the 576 remaining references for possible eligibility (see online supplementary Appendix 3) and 29 potentially

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Table 1 Strength of evidence assessment.

1	Strong evidence	Consistent findings in multiple high-quality cohort and/or case–control studies.
2	Moderate evidence	Consistent findings in multiple cohort and/or case–control studies, of which only one is a high-quality study
3	Some evidence	Findings of one cohort or case–control study or consistent findings in multiple cross-sectional studies, of which at least one study is a high-quality study
4	Inconclusive evidence	Concerns all other cases—that is, consistent findings in multiple low-quality, cross-sectional studies or inconsistent findings in multiple studies. The evidence is considered to be inconclusive if only one cross-sectional study is available, regardless of the quality of this study

Note: Findings are consistent when the results of at least 75% of the studies have similar outcomes.

relevant studies were screened in full text. After checking cross-references, two additional studies were considered potentially relevant and included.[10 19] All citations (410 in total) from the seven relevant reviews[3–9] were manually screened for potentially relevant references, which resulted in two additional references.[2 20] The experts did not provide additional references. Finally, 11 of these 33 studies assessed in full text fulfilled the criteria and were included for data extraction (see figure 1).

Study characteristics

Seven prospective cohort and four case–control studies were included. Heterogeneity of definitions of groin pain and ROM was found (see table 2).

Table 3 presents the descriptives of the 11 studies included in this review.

The prospective cohort studies lasted for one or two consecutive playing seasons. The duration of the playing seasons depended on the sport and ranged from 4 months (Australian rules football/Gaelic football) to 6 months (ice hockey) or 9 months (soccer). The results of the included studies are presented in table 4.

Methodological quality

There were 35/146 (24%) disagreements (See online supplementary appendix 4) on the risk of bias assessment, resulting in a moderate level of agreement of 0.43 (Cohen's κ).[21] All but two[22 23] included studies had a high-quality score (table 5A). The total quality score of all studies included ranged from 29% to 92%. The score of the seven prospective studies ranged from 58% to 92% and that of the four case–control studies from 29% to 71%. Three studies did not report a possible conflict of interest,[23–25] four studies reported a possible conflict[19 22 26 27] and four studies reported that no conflict of interest[28–31] was present.

The item scores for each study type are presented in table 5B. Scores <1 represent items with more negative than positive scores. These are considered as scoring relatively low.

The prospective studies had relatively low scores for items related to the standards of injury reporting, indicating the need to report how often injuries are registered during the surveillance period. This includes checking the injury criteria over time with limited interval periods (item 12). Additionally the injury criteria need to be clearly reported (ie, physical examination) (item 13).

The case–control studies perform relatively low on item (9), reporting whether or not the assessors were blinded to the disease status of the subjects and item (10) on assessing subjects before they developed symptoms or were injured. Item (16) on criteria for appropriate analysis (regression analysis) and item (17) on presenting probability data (odds) and item (18) on presenting confounding variables had relatively low scores.

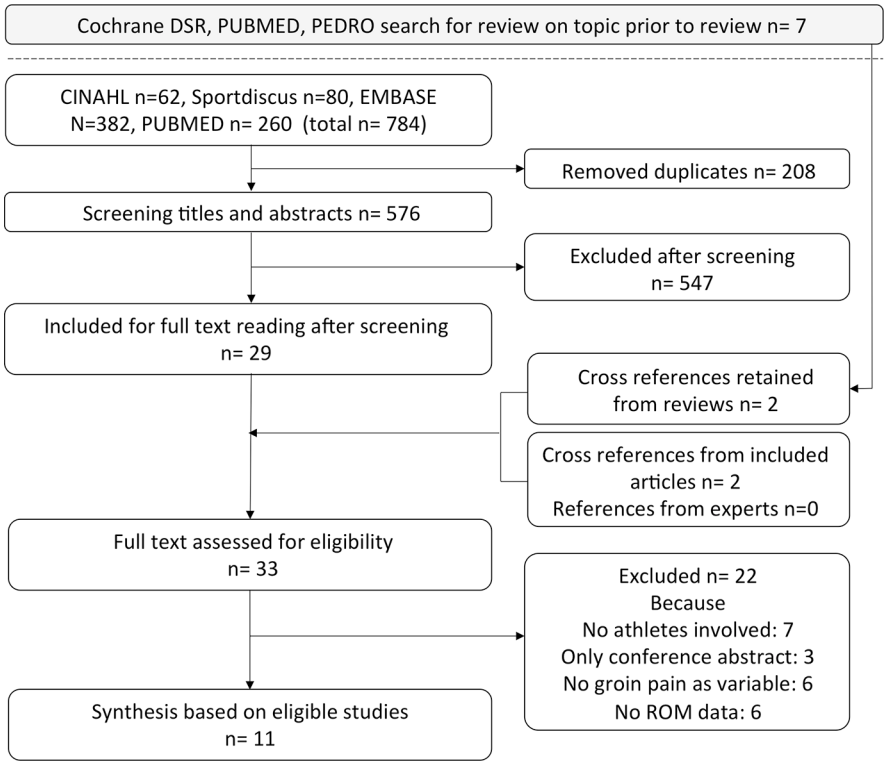

Figure 1 PRISMA flow diagram of the study search and selection procedure. DSR, database of systematic reviews; ROM, range of motion.

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Table 2 Definitions of groin pain and eligibility criteria

Groin pain definition Authors and type of study	Groin pain specified as	Pain onset (PC/CC)/pain duration (CC)	Injury time reference criterion	Exclusion of athletes at baseline
Groin strain a. Arnason *et al*[26] *(PC)* b. Engebretsen *et al*[19] *(PC)*	a. NR b. Pain on inner side of thigh	a. A+NoA b. 1. A b. 2. NoA	a. Miss one training/match b.1. Miss part of training/match b.2. No time loss	a. NR b. NR
Groin muscle strain injury Emery *et al*[27] *(PC)*	Muscle injury in the hip flexor or hip adductor muscle groups	A+NoA	Stop training/match or missed next day training/match	Groin contact injury/contusions
Adductor (muscle) strain a. Ibrahim *et al*[24] *(PC)* b. Tyler *et al*[28] *(PC)*	a+b: 1. Pain on palpation of the adductor tendons or their insertion on the pubic bone or both AND 2. Groin pain during adduction against resistance	a. NR b. A	a. Miss training/match>1 week b. Miss training/match or medical attention	a. NR b. Athletic pubalgia, osteitis pubis, hernia, hip osteoarthrosis, rectal or testicular referred pain, piriformis syndrome, pelvic or lower extremity fracture
Adductor muscle injury Witvrouw *et al*[29] *(PC)*	First time adductor muscle injury	A+NoA	Miss one training/match	Previous lower extremity muscle injury past 2 years
Chronic groin injury a. Verrall *et al* (2007)[30] *(PC)* b. Verrall *et al* (2005)[25] *(CC)*	a. 1. Pain in adductor and/or lower abdominal and/or pubic bone region AND 2. Pain for at least 6 weeks b. 1. Pain in the groin region 2. Tenderness of the pubic symphysis and/or superior pubic rami	a. NR b. NR/NR	a. Pain >6 weeks and miss one match b. Pain >6 weeks	a. Previous or current chronic groin injury b. NR
Longstanding groin pain a. Nevin *et al*[31] *(CC)* b. Malliaras *et al*[22] *(CC)*	a. 1. Pain of palpation of the adductor tendons, their insertion onto the pubic rami of pubic symphysis 2. Presence of pain during resisted adduction b. 1. Pain while running of performing rapid agility movements AND at least one of the following a. Groin pain standing on one leg b. Groin pain or stiffness in morning c. Groin pain at night d. Groin pain when coughing or sneezing	a. NR/NR b. NR/>4 weeks	a. Pain >6 weeks b. NR	a. Palpable inguinal or femoral hernia or pain felt above the conjoint tendon, clinical signs or symptoms of prostatitis or urinary tract infection, back pain felt from T10 to L5, osteoarthritis of the hip joint or clinical suspicion of a nerve entrapment syndrome b. NR
Groin pain: sports hernia Rambani[23] *(CC)*	Unilateral groin pain	NR/NR	NR	Patients with cam, pincer or combination on X-rays/MRI

Pain onset is presented as acute or non-acute for PC and CC.
Duration of pain only presented for CC.
A, acute; CC, case-control study; NoA, non-acute; NR, not reported; PC, prospective cohort study.

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Table 3 Descriptives of included studies

Authors, study type CASP score (%)	Setting Sports type/gender/level/ country	Groin pain Definition	Hip ROM Movement direction	Assessment procedure			
				Body position	Device	End range	Number of assessors/ qualification
Arnason *et al*[26] *PC, 92%*	Soccer/M/prof/Iceland	Groin strain	1. ABD 2. EXT	1. Supine, hip neutral EXT 2. Supine, hip neutral ABD (Thomas test)	Standard goniometer in degrees (°)	Standardised force assessed by tensiometer	1 *experienced PT*
Verrall *et al* [30] *PC, 92%*	ARF/M/prof/Australia	Chronic groin injury	1. IR 2. ER 3. TR	1, 2 and 3. Supine, hip and knee FL 90°	Standard goniometer in degrees (°)	Maximum range of rotation *Not further specified*	1 *NR*
Engebretsen *et al*[19] *PC, 77%*	Soccer/M/ Am/Norway	Groin strain	1. IR 2. ER 3. FL 4. EXT 5. ABD	1, 2. Supine, hip and knee FL 90° 3. Supine 4. Supine, hip neutral ABD (Thomas test) 5. Supine, hip neutral EXT	Standard goniometer in degrees (°)	According FIFA Handbook *Not further specified*	10 *Sport PTs/sport physicians*
Emery *et al*[27] *PC, 67%*	Ice hockey/M/prof/Canada	Groin muscle strain injury	1. Bilateral ABD (active)	1. Supine, hip neutral EXT, ABD performed bilaterally at once	Standard goniometer in degrees (°)	Maximum active abduction *Not further specified*	23 *PTs / physicians*
Ibrahim *et al*[24] *PC, 58%*	Soccer/M/prof/Australia	Adductor strain	1. TR	1. Supine, hip and knee FL 90°	Standard goniometer in degrees (°)	Move to end range *Not further specified*	1 *NR*
Tyler *et al*[28] *PC, 58%*	Ice hockey/M/prof/USA	Adductor muscle strain injury	1. ABD	1. Supine, hip neutral EXT	Standard goniometer in degrees (°)	Start of leg rotating externally	1 *NR*
Witvrouw *et al*[29] *PC, 58%*	Soccer/M/prof/Belgium	Adductor muscle injury	1. ABD	1. Supine, hip neutral EXT	Standard goniometer in degrees (°)	Start of leg rotating externally	2 *PT*
Nevin *et al*[31] *CC, 71%*	Gaelic football/M/Am/Ireland	Longstanding groin pain due to sport	1. IR 2. ER 3. BKFO	1. Prone, hip neutral EXT, knees together and FL 90° 2. Supine, hip neutral EXT, knees FL 90° 3. Crook-lying, hip FL 45°, knee FL 90° feet together	1,2. Inclinometer in degrees (°) 3. Rigid tape measure; distance fibula-head to plinth (cm)	1. Gentle overpressure 2. End range of motion (ROM) OR pelvis moving 3. Gentle over pressure for BKFO	1 *NR*
Verrall *et al* [25] *CC, 57%*	ARF/M/prof/Australia	Chronic groin injury	1. Bilateral IR 2. Bilateral ER 3. Bilateral TR	1, 2, 3. Supine, hip and knee FL 90°	Standard goniometer in degrees (°)	1,2,3. Maximum range of rotation *Not further specified*	1 *NR*
Malliaras *et al*[22] *CC, 47%*	ARF+soccer/M/elite junior/ Australia	Groin pain	1. IR 2. ER 3. Bilateral IR 4. Bilateral ER 5. BKFO	1. Prone, hip neutral EXT, knees FL 90° 2. Supine, hip neutral EXT, knees FL 90° 3. as per 1 4. as per 2 5. Crook-lying, hips FL 45°, knees FL 90° feet together	1, 2, 3, 4. Inclinometer in degrees (°) 5. Rigid tape measure; distance fibula-head to plinth (cm)	1. Gentle overpressure 2. End ROM OR pelvic movement 5. Gentle over pressure for BKFO	2 *NR*
Rambani *et al*[23] *CC, 29%*	NR/M+F/N/ UK	Sports hernia	1. IR 2. ER 3. FL 4. ABD 5. ADD	1, 2. Supine, hip and knee FL 90° 3. Prone 4, 5. Prone, hip neutral EXT	Standard goniometer in degrees (°)	NR	2 *NR*

Hip ROM refers to passive measures unless stated otherwise.
ABD, abduction; ADD, adduction; Am, amateur; ARF, Australian rules football; BKFO, bent knee fall out; CASP, Critical Appraisal Skills Programme; CC, case-control study; ER, external rotation; EXT, extension; F, female; FL, flexion; IR, internal rotation; M, male; NR, not reported; PC, prospective cohort study; prof, professional; PT, physical therapist; TR, total rotation (IR+ER) per hip.

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Table 4 Results of included studies

Authors, study type, CASP score (%)	Population		Groin injuries		Hip ROM	Injury analysis		Re-injury	Outcomes		Conclusion
	Non-injured players *n* *age*	Injured players *n/% age*	Injuries n	Injured hips n/total %		Per hip/per player	Data presented PS, AH or BH	Reported/statistical check or correction	Mean (SD) values for injured/non-injured hip ROM of hips (of players), p value and OR, relative risk (95% CI) with p value	Mean difference	ROM risk factor or different
Arnason *et al*[26] *PC, 92%*	294 24.0 (NR)	17/6 NR	32	13/498 3	1. ABD 2. EXT	Hips of I vs NI players	AH	Y / Y	1. 40.9 (1.1)°/43.4 (0.2)°, p=0.08. OR 0.90 (95% CI 0.80 to 1.0, p=0.05) 2. 179.0 (0.3)°/176.5 (1.4)°, p=0.14. OR/RR: NR	1. 3° 2. 3°	1. Y 2. N
Verrall *et al*[30] *PC, 92%*	29 21.4 (NR)	4/14 NR	NR	NR NR	1. IR 2. ER 3. Bilateral TR	Hips of I vs NI players	1. AH 2. AH 3. BH	N / N	1. 15.5 (2.1)/15.5 (1.3)° vs 21.8 (1.3)/20.8 (1.1)°, p=0.07/p=0.07. OR/RR:NR 2. 28.1 (1/9)/24.0 (1.4)° vs 30.7 (1.0)/29.9 (1.0)°, p=0.33/p=0.04. OR/RR:NR 3. 83.1 (4.3)°/103.2 (3.3)°, p=0.03. RR 0.90 (95% CI 0.83 to 0.99, p=0.03), robust SE 0.04, Z score −2.14	1. 6° 2. 5° 3. 20°	1. N 2. N 3. Y
Engebretsen *et al*[19] *PC, 77%*	506 23.9 (±0.2)	51/10 24.8 (0.6)	61	55/1016 5	1. IR 2. ER 3. FL 4. EXT 5. ABD	Hips of I vs NI players	PS	N / N	1. 27.5 (2.0)°/29.8 (0.5)°, p=NR. OR 1.06 (95% CI 0.77 to 1.44, p=0.73) 2. 42.3 (1.8)°/46.3 (0.4)°, p=NR. OR=1.53 (95% CI 1.13 to 2.07, p<0.01) 3. 123.7 (1.5)°/120.8 (0.5)°, p=NR. OR=0.95 (95% CI 0.71 to 1.28, p=0.74) 4. 19.7 (1.1)°/20.9 (0.3)°, p=NR. OR 1.15 (95% CI 0.85 to 1.55, p=0.37) 5. 52.1 (1.6)°/51.0 (0.4)°, p=NR. OR 0.95 (95% CI 0.70 to 1.28, p=0.73)	1. 2° 2. 4° 3. 3° 4. 1° 5. 1°	1. N 2. Y 3. N 4. N 5. N
Emery *et al*[27] *PC, 67%*	a. 1292 (camp) b. 647 (season) NR	NR NR	a. 52 b. 152	NR NR	1. Bilateral ABD (active)	Hips of I vs NI players	BH	Y / Y	1a. Camp: abduction <67.5°; RR 0.56 (95% CI 0.18 to 1.76, p=NR) 1b. Season: abduction <67.5°; RR 0.48 (95% CI 0.15 to 1.53, p=NR)	-	1a. N 1b. N
Ibrahim *et al*[24] *PC, 58%*	101 NR	8/8 NR	9	NR NR	1. TR	a. D hips of I vs NI players b. ND hips of I vs NI players c. I hips vs NI hips in I players	a. PS b. PS c. AH	Y / N	1a+b. 44.7 (NR)°/53.7 (NR)°, (a. p=0.03; b. hip p=0.04). OR/RR: NR 1c. NR, p=0.06. OR/RR: NR	a. 9° b. 9° c. 3°	1a. Y 1b. Y 1c. N
Tyler *et al*[28] *PC, 58%*	47/17 (1/2 seasons FU) 23.0 (±4.3)	8/17	11	NR NR	1. ABD	a. Hips of I vs NI players b. I hips vs NI hips	a. AH b. PS	Y / Y	1a. (46.3 (10.3)°/45.8 (11.0)°, p=0.92). OR/RR: NR 1b. (46.3 (10.3)°/46.3 (9.2)°, p=0.69). OR/RR: NR	1a. 2° 1b. 0°	1a. N 1b. N
Witvrouw *et al*[29] *PC, 58%*	146	NR	13	NR	1. ABD	Hips of I vs NI players	AH	N / N	1. (51.3(NR)°/52.3(NR)°, p=0.45). OR/RR: NR	1. 1°	1. N

Continued

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Table 4 Continued

Authors, study type, CASP score (%)	Population		Groin injuries		Hip ROM	Injury analysis		Re-injury	Outcomes		Conclusion
	Non-injured players *n* *age*	Injured players *n/%* *age*	Injuries n	Injured hips n/total %		Per hip/per player	Data presented PS, AH or BH	Reported/statistical check or correction	Mean (SD) values for injured/non-injured hip ROM of hips (of players), p value and OR, relative risk (95% CI) with p value	Mean difference	ROM risk factor or different
Nevin *et al*[31] *CC, 71%*	18 23.9 (±3.2)	18 23.8 (±3.6)	NA	22	1. IR 2. ER 3. BKFO	a. I hips I players vs D hips of NI players b. NI hips I players vs ND hips NI players	PS	N / N	1a. 30.6 (4.9)°/34.5 (5.6)°, p<0.05 1b. 28.6 (6.1)°/34.5 (5.9)°, p<0.05 2a. 27.2 (5.3)°/29.9 (4.5)°, p<0.05 2b. 26.5 (5.7)°/30.3 (4.7)°, p<0.05) 3a. 19.3 (4.3)cm/15.1 (3.8)cm, p<0.05 3b. 18.7 (3.4)cm/15.2 (4.3)cm, p<0.05	1a. 4° 1b. 6° 2a. 3° 2b. 4° 3a. 4 cm 3b. 4 cm	1a. Y 1b. Y 2a. Y 2b. Y 3a. Y 3b. Y
Verrall *et al*[25] *Case–control* *57%*	42 NR	47 NR	NA	NR	1. Bilateral IR 2. Bilateral ER 3. Bilateral TR	Hips of I vs NI players	1, 2, 3 BH	Y / N	1. 36.7 (10.1)°/41.4 (11.3)°, p=0.03 2. 55.2 (10.5)°/60.3 (9.2)°, p=0.01 3. 91.3 (17.6)°/101 (17.1)°, p<0.01	1. 5° 2. 5° 3. 10°	1. Y 2. Y 3. Y
Malliaras *et al*[22] *Case–control* 47%	19 17.1 (±1.6)	10 17.3 (±0.8)	NA	18	1. IR 2. ER 3. Bilateral IR 4. Bilateral ER 5. BKFO	L+R hips of I vs NI players	PS	Y/N	1. R 34.4 (8.1)/34.2 (11.6), p=0.96, L 32.7 (10.7)/33.3 (13.6), p=0.91 2. R 39.4 (8.7)/40.8 (7.1), p=0.66, L 46.2 (13.4)/48.3 (9.6), p=0.63 3. Not different, p=0.96 4. Not different, p=0.61 5. R 16.4 (4.8)cm/12.9 (5.1)cm, p=0.09, L 14.4 (3.8)/12.5 (4.7), p=0.28	1. R 0°/L 1° 2. R 1.5°/L 2° 3. NR 4. NR 5. R 4 cm/L 2 cm	1. N 2. N 3. N 4. N 5. N
Rambani *et al*[23] *Case–control* *29%*	25 (18M/7F) 23.2(NR)	25 (18M/7F) 22.8 (NR)	NA	25	1. IR 2. ER 3. FL 4. ABD 5. ADD	Hips of I vs NI players	AH	N	1. 17.4 (NR)°/40.9 (NR)°, p=0.03 2. 26.2 (NR)°/45.1 (NR)°, p=0.03 3. 122.3 (NR)°/122.2 (NR)°, p=NR 4. 35.2 (NR)°/35.9 (NR)°, p=NR 5. 27.1 (NR)°/26.20 (NR)°, p=NR	1. 24° 2. 19° 3. 0° 4. 1° 5. 1°	1. Y 2. Y 3. N 4. N 5. N

Hip ROM refers to passive measures unless stated otherwise. Re-injury: a re-injury of one player during the study. Age presented as mean (SD)
Data presentation per side (PS) whether left(L)/right(R) or dominant(D)/non-dominant(ND), for all hips (AH) or both hips (BH) as sum scores of left and right.
ABD, abduction; ADD, adduction; BKFO, bent knee fall out; CASP, Critical Appraisal Skills Programme; CC, case-control study; corr, correction; ER, external rotation; EXT, extension; F, female; FL, flexion; I, injured; IR, internal rotation; M, male; N, no; NI, not injured; NR, not reported; OR, odd's ratio; PC, prospective cohort study; RR, relative risk; TR, total rotation (IR+ER) per hip; Y, yes.

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Table 5 Quality assessment of included studies according to the CASP criteria checklist and CASP ratio positive score/negative score

A: Methodological quality

	Study type	1. Clear study objective (CS, CC, PC)	2. Population charac-teristics (CS, CC, PC)	3. Cases/ controls same population (CC)	4. Partici-pation rate (CS, CC, PC)	5. Partici-pation rate followup (PC)	6. Data presented properly (CS, CC, PC)	7. Instru-mented hip ROM assessment (CS, CC, PC)	8. Hip ROM assess-ment injured/ controls (CC)	9. Hip ROM assess-ment blinded injured/ controls (CS, CC)	10. Hip ROM assessed before onset of groinpain (CC)	11. Data collection for ≥1playing season (PC)	12. Data collection at least every 3 months (PC)	13. Physical examina-tion blinded to exposure status (CS, CC, PC)	14. Prospective enrolment of subjects (CC)	15. Uni- ormulti-variate analysis performed (CS, PC)	16. Logistic regression analysis performed (CC)	17. Relation measures (OR/ RR) with 95% CI (CS, CC, PC)	18. Confounding / effect modification controlled (CS, CC, PC)	19. N cases is 10 times N independent variables (at least) (CS,CC,PC)	total	%
Arnason *et al*[26]	PC	+	+	NA	+	+	+	+	NA	NA	NA	+	+	–	NA	+	NA	+	+	+	12/13	92%
Verrall *et al* 2007[30]	PC	+	+	NA	+	+	+	+	NA	NA	NA	+	–	+	NA	+	NA	+	+	+	12/13	92%
Enge bretsen *et al*[19]	PC	+	+	NA	+	+	+	–	NA	NA	NA	+	–	–	NA	+	NA	+	+	+	10/13	77%
Emery *et al*[49]	PC	+	–	NA	+	–	+	+	NA	NA	NA	+	–	–	NA	+	NA	+	+	NA	8/12	67%
Ibrahim *et al*[24]	PC	+	–	NA	+	+	–	+	NA	NA	NA	+	+	+	NA	–	NA	–	–	NA	7/12	58%
Tyler *et al*[28]	PC	+	+	NA	–	–	+	+	NA	NA	NA	+	–	+	NA	+	NA	–	–	NA	7/12	58%
Witv rouw *et al*[34]	PC	+	+	NA	–	–	+	+	NA	NA	NA	+	+	–	NA	+	NA	–	–	NA	7/12	58%
Nevin *et al*[31]	CC	+	+	+	+	NA	+	+	+	–	–	NA	NA	+	+	NA	–	–	+	NA	10/14	71%
Verrall *et al* 2005[25]	CC	+	–	+	+	NA	+	+	+	+	–	NA	NA	+	–	NA	–	–	–	NA	8/14	57%
Malliaras *et al*[22]	CC	+	+	–	–	NA	+	+	+	–	–	NA	NA	+	–	+	–	–	–	NA	7/15	47%
Rambani *et al*[23]	CC	–	+	–	–	NA	–	+	+	–	–	NA	NA	–	+	NA	–	–	–	NA	4/14	29%
B: CASP ratio																						
Relative CASP quality score	PC	7/0	5/2	NA	5/2	4/3	6/1	6/1	NA	NA	NA	7/0	**3/4**	**3/4**	NA	6/1	NA	4/3	4/3	3/0		
	CC	3/1	3/1	2/2	2/2	NA	3/1	4/0	4/0	**1/3**	**0/4**	NA	NA	3/1	2/2	NA	**0/4**	**0/4**	**1/3**	NA		

A: Methodological quality scores with the total quality score for all positive validity/precision items and the percentage of the maximum attainable score (%).
B: CASP ratio scores <1 (more negative than positive scores per item for all studies) are in bold.
CASP, Critical Appraisal Skills Programme; CC, case–control study; CS, cross-sectional study; NA, not applicable item for that study type; PC, prospective cohort study; ROM, range of motion; '+', positive score for item criteria; '–', negative score for item criteria.

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Table 6 Strength of evidence assessment whether ('for') or not ('against') all identified hip ROM measures have a relationship (risk of differentiating) with groin pain in athletes. (Findings are presented per hip unless otherwise stated.)

Strength of evidence	Findings
Strong	*Prospective RF studies:* For: lower TR of both hips (BH) (SHKF)[24 30] Against: hip IR (SHKF),[19 30] abduction[19 26–29] and extension[19 26]
Moderate	*Differentiates asymptomatic players from symptomatic ones:* For: lower IR,[23 25] ER[23 25] of BH (SHKF)
Some	*Prospective RF studies:* Against: TR per hip (SHKF),[24] hip flexion[19] and bilateral abduction[27] *Differentiates asymptomatic players from symptomatic ones:* For: lower TR of BH (SHKF)[25] Against: combined (bilateral) hip IR (PHNE) and ER (SHNE)[22]
Inconclusive	*Prospective RF studies:* Hip ER (SHKF)[19 30] *Differentiates asymptomatic players from symptomatic ones:* Hip IR (PHNE),[22 31] ER (SHNE)[22 31] and BKFO[22 31]
Not studied or single low quality study	*Prospective RF studies:* Hip IR, ER, TR (all with neutral hip extension), BKFO and adduction *Differentiates asymptomatic players from symptomatic ones:* Hip extension, flexion, abduction and adduction.

BKFO, bent knee fall out; ER, external rotation; IR, internal rotation; PHNE, prone hip neutral extension; RF, risk factor; ROM, range of movement; SHKF, supine position with hip and knee flexed to 90°; SHNE, supine hip neutral extension; TR, total rotation.

Overall strength of evidence

Table 6 shows the strength of evidence for all ROM measures as risk factors, identified by prospective studies, or as differentiating factors as identified by case–control studies.

DISCUSSION

This systematic review included seven high quality prospective and four case–control studies (two high and two low quality) on the relationship between groin pain in athletes and hip ROM. Heterogeneous definitions for groin pain/injury and measurement techniques for hip ROM prevented data pooling. A strength-of-evidence synthesis showed that smaller (<85°) total rotational hip ROM (hips and knees 90° flexed) is the most consistent risk factor for development of groin pain and differentiates athletes with groin pain from those without. However, the number of studies with homogeneous methods (two and one, respectively) is limited.

Heterogeneity in terminology and definitions

Heterogeneous terminology and definitions, as identified in the included studies, has also been highlighted in a Delphi study among groin pain experts.[32] There was also large variability in the injury and time-loss criteria, reflecting different stages of the injury spectrum.[33] This ranged from inclusion of players with pain who were still able to play[19] to players with a first-time groin time-loss injury[34] or those with time loss and pain >6 weeks.[30] The magnitude of time loss ranged considerably from missing part of one match to missing >1 week of play.[24] Altering thresholds for injury definition affects injury prevalence and incidence, which will influence the results and interpretation of the study findings.

ROM measures

There are important concerns about heterogeneity and measurement properties to consider when interpreting these data.

Heterogeneity

Even though 11 studies report hip ROM and groin pain, the strength-of-evidence assessment, indicating total rotational ROM measure as being related to groin pain, was based on only two studies. Other studies could not be included in this synthesis, owing to heterogeneous hip ROM measures, different assessment methods[35] and whether or not the analysis was performed for each hip[19 22 24 28 30 31] or for combined (either internal rotation (IR) or external rotation (ER)[25 30] or for bilateral[23–30]) ROM values.

Validity and reliability of hip ROM measures

When measuring rotational hip ROM, the validity will be influenced by the measurement protocol. One specific element is the criterion used to determine end ROM. Overestimation is possible if the measurements are not conducted in a single plane, or if compensation from the pelvis and lower back are permitted.[35] End ROM criteria are generally under-reported,[19 23–25 27 30] which limits assessment of study validity.

A large number of raters (a range of 1–23 raters was found) will result in increased rater variability, and this reduces measurement reliability,[36 37] and increases the standard error of measurement (SEM) and minimal detectable change (MDC)[22] values for hip ROM measures. Using a single examiner can deal with this, but this limits extrapolation of the findings to different settings. Electronic assessments of forces applied may be useful to increase reliability,[38] but this was performed in only one of the studies.[26] Reduction of compensatory movements and applying standardised force with a specially developed examination chair has been reported to strongly increase the reliability of hip ROM assessment.[39]

Minimal detectable change

To detect true differences or changes in ROM—that is, changes that are not the result of measurement error, differences must at least equal or exceed the MDC. No studies in this review dealt with this topic. For IR and ER (hips and knees 90° flexed) assessed with a goniometer, the SEM (intra-rater) was previously reported to be 2° and 3°.[38] Computing the MDC, based on SEM values according the formula MDC = $(1.96 \times SEM \times \sqrt{2})$, reveals MDCs for both IR and ER of 7°. For assessment with hips in neutral extension the SEM (intra-rater) is 2° and 4°,[22] resulting in MDC for IR of 6° and ER of 11°. The differences found for rotational ROM often exceed the SEM, but not the MDC. Thus, the available evidence suggests that ROM is not an appropriate screening measure for predicting groin pain. Future studies could combine data from larger cohorts to improve the accuracy of the estimation of risk. However, to pool such studies, homogeneity of the items previously discussed in this paper is a prerequisite.

What to consider when performing new studies?

We suggest that future studies should consider examining and reporting all ROM measures available for each hip either unilaterally or bilaterally and for BH (combined). Appropriate stabilisation of the pelvis, a clear definition of end ROM and forces

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applied should be used and described in detail. The number of raters should be as low as possible.

ROM and groin pain

The evidence from this review does not support a relationship between hip ROM and groin pain, which is at odds with clinical perceptions. It has been hypothesised that restricted hip rotations induce increased stress over the symphysis and surrounding soft tissues.[10 25 40] One cadaver study[11] observed that the presence of cam morphology, which is associated with decreased internal rotation,[41] increased the shear forces and rotational movement of the symphysis. Yet, no in vivo validation has been performed of the effects of lower hip ROM measures. A recent paper, published after the search period of this review, showed that hip ROM, when assessed in a sport-specific way, is lower on the injured side in players with unilateral longstanding adductor-related groin pain.[42] It was postulated that this negatively affects biomechanical characteristics[42] and hinders adequate energy transfer between body segments[43] during sporting tasks, resulting in supraphysiological tissue loading.

Confounders

Previous injury is a risk factor for re-injury.[3] Two prospective studies accounted for this by recording re-injuries[29 30] or correcting for their presence in the analyses, whereas others did not.[19 24 26 28]

Age is a risk factor for groin pain.[3] Some prospective studies reported participants' ages for the studied[19 26–28] and injured[19 27] populations and accounted for age in the analyses,[26 27 30] and two[19 30] corrected for this association when indicated. However, others do not report age.[24 29] ROM is related to age and has also been shown to decrease in older players and in those playing at higher levels.[44]

When ER and/or IR are studied in isolation, they are not risk factors for the development of groin pain. However, when combined (summated) to give total rotation, there is an increased risk when total rotation is lower. Individual anatomical differences between version of the femoral neck, resulting in greater or smaller ER or IR, may be confounders.[45]

Smaller internal rotation has been associated with the presence of cam morphology.[41] This morphological appearance may be a confounder as it is highly prevalent in athletes, but the reported prevalence estimates have a high risk of bias.[46]

Therefore, future studies should report and correct for previous injury, age and presence of cam morphology, as these are recognised confounders for groin pain and ROM.

LIMITATIONS

We identified many factors that might influence the outcome of the selected studies, such as reporting and consistency of measurement techniques, which were generally poorly or not reported at all. There were also many differences in the definitions of 'groin pain' and 'injury'. This combination of poor reporting and use of heterogeneous injury definitions and measurement methods prevented a comparison of many study findings. This limited our ability to group studies to provide higher levels of evidence.

When consulting the current strength of evidence assessment, 'strong evidence' may be an 'overqualification' in our review. The strength of evidence method used means that having two high-quality studies qualifies as 'strong evidence'. For some readers 'strong evidence' may intuitively refer to a larger body of evidence. A general limitation threatening the validity of this, like any other review, is that non-published data may exist together with publication bias. A study reporting negative findings—that is, no relationship between ROM and groin pain, may be less likely to be published. We used SPORTDiscus and contacted experts in the field in an effort to find other papers. A more comprehensive grey literature search might have helped. All but one of the studies were on male athletes and thus, information from this review cannot be extrapolated to female athletes, leaving space for new research.

When you are a clinician, seeing patients with groin pain you may consider

- Single-observer ROM assessment, when performed with measurement devices, can detect changes in hip ROM over time. However, the changes may only be true if they exceed 7° for either IR or ER (hip and knee flexed).
- Considering that total rotational ROM of both hips is lower in athletes with groin pain, improving it as part of treatment should be considered. However, as the differences found are generally small, this should not be the only intervention. It is also difficult to identify which patients may benefit.
- Screening for hip ROM to prevent groin injury is unlikely to detect an athlete at risk. When a large deficit is found, prevention can be considered, although what this should entail remains unclear.

When you perform research on hip ROM in athletes with groin pain and want to contribute to the existing knowledge you should consider

- Using clear and generally accepted injury definitions and terminology;[1 47 48]
- Reporting the physical examination findings of injury and ROM measurement techniques comprehensively (eg, both IR, ER and summated measures) and providing data on measurement properties (eg, SEM and MDC);
- Blinding of assessors for participant's injury status and injured side(s);
- Presenting risk estimates (absolute risk/OR/risk ratio), with and without confounding variables.

CONCLUSION

There is strong evidence that lower total hip ROM of both hips is a risk factor for the development of groin pain. There is strong evidence that internal rotation, abduction and extension are not risk factors for the development of groin pain. Screening for hip ROM is unlikely to correctly identify an athlete at risk because of the small ROM differences found, which were lower than the known measurement errors.

What is already known on this topic and what this study adds

- No information exists on how to apply the available information on hip range of motion (ROM) and groin pain in athletes.
- There is strong evidence supporting the statement that smaller total rotation (<85°) of both hips at a pre-season screening increases the risk of developing groin pain other ROM determinants do not.
- Hip ROM assessment does not allow correct identification of an athlete at risk of groin pain.

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Competing interests None declared.

Provenance and peer review Not commissioned; externally peer reviewed.

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